Housing Price Prediction using Linear Regression

Objective

Implement and understand Simple and Multiple Linear Regression using Scikit-learn to predict house prices.

Tools & Libraries

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-learn

Dataset

Housing Dataset containing various house features and their corresponding prices.

Steps Performed

1. Imported and explored the dataset.
2. Preprocessed the data and handled categorical variables.
3. Split the dataset into training and testing sets.
4. Trained a Linear Regression model.
5. Predicted house prices on test data.
6. Evaluated model performance using:
7. Mean Absolute Error (MAE)
8. Mean Squared Error (MSE)
9. R^2 Score
10. Visualized Actual vs Predicted Prices.
11. Analyzed feature coefficients.

Results

The Linear Regression model was successfully trained and evaluated. Performance metrics such as MAE, MSE, and R^2 Score were used to measure prediction accuracy.

Files

- Housing(1).csv

- Task3_Linear_Regression.ipynb
- README.md

Conclusion

Linear Regression is an effective supervised learning algorithm for predicting continuous values such as house prices. This project demonstrates the complete workflow from data preprocessing to model evaluation.